

MATAM PADMINI

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EDUCATION

Final Year (8th Semester)

- **B.E. in Artificial Intelligence and Data Science** **2022-2026**
Stanley College of Engineering and Technology for Women | CGPA: 9.33
- **Intermediate (MPC)** **2020-2022**
Sri Chaitanya Junior College | CGPA: 9.68
- **SSC** **YOP: 2020**
Sri Sai Krishna Talent School | CGPA: 10.0

EXPERIENCE

Operations Associate Intern | Deccan AI

October 2025 - Present

- Gained hands-on experience with RLHF pipelines, including complex prompting, evaluation, and annotation of LLM outputs to improve model quality and data accuracy.
- Actively contributing to internal operations by assisting with task coordination, workflow management, and maintaining annotation consistency across projects.

Research Intern | Indian Institute of Technology, Hyderabad

May 2025 – August 2025

- Worked on “Air Pollution and Climate Change: Modeling and Analysis,” handling raw CSV climate datasets for Delhi and Chennai stations, performing preprocessing, EDA, and building predictive models using Python.
- Contributed to climate research insights through data analysis, visualization, and project documentation under faculty mentorship.

AI/ML Intern | InternPe

April 2024 – May 2024

- Worked on AI/ML projects including Diabetes Prediction, Car Price Prediction, IPL Winner Forecasting, and Breast Cancer Detection using Python, Pandas, NumPy, and Scikit-learn.

PROJECTS

SFlowGCN: Low-Rate Flow Table Overflow Attack Detection in SDN

January 2026 – April 2026

- Built an end-to-end SDN emulation testbed using Mininet, Ryu Controller, and Open vSwitch with sFlow packet sampling, capturing, and processing 1.17M+ packet records across six diverse LFTO attack configurations.
- Engineered a 15-dimensional node feature vector including 6 novel LFTO-specific features: Burst Score, Protocol Entropy, IAT Skewness, UDP Ratio, ICMP Ratio, and Port Diversity, targeting stealthy periodic injection patterns.
- Constructed PaFlo-Graphs using 200-packet sliding windows (5,530 balanced graphs) and implemented SFlowGCN with 5 GCN layers, BatchNorm, residual connections, triple pooling (mean+max+add), attention gate, and Stochastic Weight Averaging using PyTorch Geometric.
- Achieved 96.47% accuracy, 99.69% ROC-AUC, and 95.30% TPR; live deployment reduced flow table overflow events by 83.4% and Packet-In messages by 63.6% with 0.005ms average GCN inference latency.

QA Bot

February 2025 – May 2025

- Built an RAG-based Question-Answering bot using Python, LangChain, IBM Watsonx LLMs, and ChromaDB to retrieve contextually accurate answers from PDF documents, achieving 97% accuracy. Developed a Gradio interface for easy user interaction and demonstrated applications in research, education, and enterprise knowledge management.

Hackathons & Competitions

ACM-W HACKATHON (3rd Place)

November 2024

- Developed a web-based career recommendation system using HTML, CSS, and basic JavaScript logic to suggest career paths based on user interests and skills.

GDSC HACKATHON

February 2024

- Built a CNN model with transfer learning to classify metastatic cancer detection in pathology images using the PCAM dataset, achieving 90% accuracy in a hackathon with 50+ teams. Utilized TensorFlow, OpenCV, and Python for image preprocessing and model training.

SKILLS

Languages: Python, Java (Intermediate), C, SQL, R (Basics)

ML / DL: Scikit-learn, TensorFlow, PyTorch, PyTorch Geometric, NetworkX

SDN / Networking: Mininet, Ryu SDN Controller, Open vSwitch, sFlow, hping3, iperf3

Data & Tools: Pandas, NumPy, Matplotlib, LangChain, ChromaDB, Jupyter, GitHub, VS Code

Concepts: Graph Neural Networks, Deep Learning, Machine Learning, Data Analytics, NLP, Computer Network, DBMS, OOP

Basic knowledge in: Web Development, Web Analytics, MongoDB, NLP, Compiler Design

CERTIFICATIONS

Google Data Analytics Professional (Coursera) | AI Fundamentals (IBM) | Web Development Basics (IBM) | OOPS using Java & SQL for Data Science (Great Learning) | C Training (IIT Bombay Spoken Tutorial) | AI Fluency: Framework and Foundations (Anthropic) | WE-Hub IDEATHON-2.0 Idea Validation